

Testing Video Transmission in Ns-2 Using the EvalVid Framework

Ns-2 is a discrete event simulator targeted at networking research. It provides support for simulation of TCP, routing and multicast protocols over wired and wireless (local and satellite) networks.

Conventionally, the Internet has been used for data applications like email, chat, Web surfing, etc. Multimedia technology has been around for quite some time, but with the pace of development that it has enjoyed in the last few years, it has become an integral part of almost every Web endeavour. It finds application in fields ranging from video lectures to catalogues and games to video clips. The combination of multimedia technology and the Internet is currently attracting a considerable amount of attention from researchers, developers and end users.

Video transmission nowadays has become a necessity for almost every Internet user, irrespective of the device used to access the Internet. Typically, video transmission demands high bandwidth. Without compression, it is very difficult to transmit video over wired or wireless networks.

With the increasing use of multimedia-capturing devices and the improved quality of capture, the demand for high quality multimedia has gone up. The traditional protocols used to carry such volumes of multimedia data have started showing some limitations. Hence, enhanced protocols are the need of the hour. Researchers working on developing such protocols need platforms to test the performance of these newly designed protocols. The performance of the suite of protocols should ideally be measured on a real network. But due to several limitations of real networks, simulation platforms are commonly used to test prototypes.

This article is aimed at demonstrating the widely used and open source simulation platform, ns-2, for measuring the performance of protocols when multimedia content is transmitted over the network.

Architecture of ns-2 and EvalVid

Among researchers, network simulator 2 (ns-2) is one of the most widely used, maintained and trusted simulators. It is useful for simulating a wide variety of networks, ranging from



Ethernet and wireless LANs, to ad hoc networks and wide area networks. The architecture of ns-2 is shown in Figure 1.

Conventionally, ns-2 supports constant bit rate and traffic in bursts, which resembles most of the data applications on the Internet. Multimedia traffic has characteristics that are different from data traffic. Data traffic is mostly immune to jitter and less sensitive to delay, whereas multimedia applications demand a high quality of service. Generating traffic representing multimedia data is very important when evaluating the performance of any network.

EvalVid is a framework and tool-set for evaluating the quality of video transmitted over a physical or simulated communication network. In addition to measuring the QoS (Quality of Service) parameters of the underlying network, like delays, loss rates and jitter, it also provides support for subjective video quality evaluation of the received video, based on frame-by-frame PSNR (Peak Signal to Noise Ratio)